

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

Which SQL constraint ensures that a column cannot contain NULL values?

- A) UNIQUE
- B) NOT NULL
- C) CHECK
- D) PRIMARY KEY

Answer: B

Question 2

Given the table:

```
CREATE TABLE Employee (  
  empID INT PRIMARY KEY,  
  name VARCHAR(50),  
  salary DECIMAL(10,2) CHECK (salary > 20000),  
  deptID INT DEFAULT 10  
);
```

What happens when you execute: INSERT INTO Employee (empID, name, salary) VALUES (101, 'John', 15000);

- A) Row is inserted successfully
- B) Error: CHECK constraint violation
- C) Row inserted with salary set to NULL
- D) Row inserted with salary set to 20000

Answer: B

Question 3

What is the difference between DELETE and DROP commands?

- A) DELETE removes table structure, DROP removes rows
- B) DELETE removes rows, DROP removes the entire table including structure
- C) They are synonyms

- D) DELETE is DDL, DROP is DML

Answer: B

Question 4

Consider: SELECT name FROM student WHERE name LIKE 'A_i%'; Which names match this pattern?

- A) Ajith, Aruni, Anil
- B) Ali, Amin, Aruni
- C) Ajith, Ali, Amin
- D) Anil, Aruni only

Answer: C

Question 5

What does this query return?

SELECT COUNT(*) FROM student WHERE gpa IS NULL;

- A) Number of students with gpa = 0
- B) Number of students without gpa recorded
- C) Total number of students
- D) Error: cannot count NULL values

Answer: B

Question 6

Given Student(sid, name, gpa, progId) and Program(progId, name, duration), which query finds students in 4-year programs?

- A) SELECT s.name FROM Student s, Program p WHERE duration = 4;
- B) SELECT s.name FROM Student s WHERE s.progId IN (SELECT progId FROM Program WHERE duration = 4);
- C) SELECT name FROM Student WHERE duration = 4;

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D) SELECT s.name FROM Student s JOIN Program p WHERE p.duration = 4;

Answer: B

Question 7

What is the result of:

SELECT DISTINCT progId FROM student; compared
to:

SELECT progId FROM student;

- A) Same result always
- B) DISTINCT removes duplicate program IDs
- C) DISTINCT sorts the results
- D) DISTINCT shows only NULL values

Answer: B

Question 8

Which query finds students with gpa between 3.0 and 3.5 (inclusive)?

- A) SELECT * FROM student WHERE gpa > 3.0 AND gpa < 3.5;
- B) SELECT * FROM student WHERE gpa >= 3.0 AND gpa <= 3.5;
- C) SELECT * FROM student WHERE gpa BETWEEN 3.0 AND 3.5;
- D) Both B and C

Answer: D

Question 9

What does ORDER BY do without ASC or DESC specification?

- A) Orders randomly
- B) Orders in ascending order (default)
- C) Orders in descending order
- D) Produces error

Answer: B

Question 10 Given:

SELECT s.name, p.offerBy

FROM Student s LEFT OUTER JOIN Program p ON s.progId = p.progId; What happens to students whose progId doesn't match any Program?

- A) They are excluded from results
- B) They appear with offerBy = NULL
- C) They appear with offerBy = 'Unknown'
- D) Query produces error

Answer: B

Question 11

Which is correct about INNER JOIN vs LEFT OUTER JOIN?

- ☐ A) INNER JOIN returns more rows
- B) LEFT OUTER JOIN returns all rows from left table
- C) They always produce same result
- D) INNER JOIN includes NULL values

Answer: B

Question 12 Consider:

SELECT name FROM Student

WHERE sid IN (SELECT sid FROM Grades WHERE grade = 'A');

This is equivalent to:

- A) SELECT DISTINCT s.name FROM Student s, Grades g WHERE s.sid = g.sid AND g.grade = 'A';
- B) SELECT s.name FROM Student s WHERE EXISTS (SELECT * FROM Grades g WHERE g.sid = s.sid AND g.grade = 'A');

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C) Both A and B

D) Neither A nor B

Answer: C

Question 13

What's the difference between ANY and ALL operators?

- A) ANY requires all comparisons true, ALL requires at least one true
- B) ANY requires at least one comparison true, ALL requires all comparisons true
- C) They are synonyms
- D) ANY is for subqueries, ALL is for joins

Answer: B

Question 14

Given Student(sid, name, gpa), which query finds students with gpa greater than ALL Computer Science students?

- A) SELECT name FROM Student WHERE gpa > (SELECT MAX(gpa) FROM Student WHERE major = 'CS');
- B) SELECT name FROM Student WHERE gpa > ALL (SELECT gpa FROM Student WHERE major = 'CS');
- C) Both A and B
- D) Neither works

Answer: C

Question 15

What does COUNT(columnName) do differently from COUNT(*)?

- A) They are identical
 - B) COUNT(columnName) excludes NULL values in that column
 - C) COUNT(*) excludes NULL values
 - D) COUNT(columnName) counts distinct values **Answer: B**
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Question 16

In GROUP BY clause, which is TRUE?

- A) Can only use one grouping attribute
- B) All SELECT attributes must appear in GROUP BY or be aggregated
- C) Cannot use WHERE with GROUP BY
- D) Must always use HAVING

Answer: B

Question 17

What's the difference between WHERE and HAVING clauses?

- A) WHERE is used with GROUP BY, HAVING is not
- B) WHERE filters rows before grouping, HAVING filters groups after grouping
- C) They are synonyms
- D) HAVING is used with aggregate functions only **Answer: B**

Question 18 Consider:

SELECT deptID, AVG(salary)

FROM Employee

WHERE salary > 30000 GROUP

BY deptID

HAVING COUNT(*) > 5;

What does this query return?

- A) Departments with average salary > 30000 and more than 5 employees
- B) Average salary of departments having more than 5 employees earning over 30000
- C) All departments with more than 5 employees
- D) Error: cannot use WHERE with GROUP BY **Answer: B**

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Question 19

What constraint type is violated if you try to insert duplicate values in a PRIMARY KEY column?

- A) NOT NULL constraint
- B) CHECK constraint
- C) UNIQUE constraint (implicit in PRIMARY KEY)
- D) FOREIGN KEY constraint

Answer: C

Question 20 Given:

ALTER TABLE Student ADD CONSTRAINT chk_age CHECK (age >= 18); What happens to existing rows with age < 18?

- A) They are automatically deleted
- B) They are updated to age = 18
- C) The ALTER statement fails
- D) They remain but new inserts are checked

Answer: C

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply)**Question 21**

Which are valid SQL aggregate functions? **(Select all that apply)**

- A) SUM()
- B) AVG()

- C) MEDIAN()
- D) COUNT()
- E) MAX()
- F) MIN()

Answers: A, B, D, E, F

Question 22

Which constraints can be defined at the column level in CREATE TABLE? **(Select all that apply)**

- A) NOT NULL
- B) UNIQUE
- C) PRIMARY KEY
- D) FOREIGN KEY
- E) CHECK
- F) DEFAULT

Answers: A, B, C, D, E, F

Question 23

For ON DELETE CASCADE in a foreign key constraint: **(Select all that apply)**

- A) Deleting parent row automatically deletes child rows
- B) Maintains referential integrity automatically
- C) Prevents deletion of parent rows
- D) Is safer than ON DELETE SET NULL
- E) Propagates deletions to dependent tables

Answers: A, B, E

Question 24

Which statements about DISTINCT are TRUE? **(Select all that apply)**

- A) Eliminates duplicate rows from result set
- B) Can be used with aggregate functions
- C) Affects performance (requires sorting/hashing)
- D) Can only be used with single column
- E) DISTINCT NULL values are treated as duplicates

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Answers: A, B, C, E

Question 25

For the query:

```
SELECT deptID, COUNT(*), AVG(salary)
```

```
FROM Employee
```

```
GROUP BY deptID;
```

Which are TRUE? **(Select all that apply)**

- A) Returns one row per department
- B) COUNT(*) counts employees per department
- C) AVG(salary) shows average salary per department
- D) Can add ORDER BY deptID
- E) Can add WHERE salary > 50000 before GROUP BY

Answers: A, B, C, D, E

Question 26

Which operations require a subquery? **(Select all that apply)**

- A) IN operator with dynamic value list
- B) EXISTS operator
- C) Comparing to aggregate results from different groups
- D) ALL operator
- E) Simple INNER JOIN

Answers: A, B, C, D

Question 27

For LEFT OUTER JOIN between Student and Program: **(Select all that apply)**

- A) All Student rows appear in result
- B) Program columns are NULL for students without matching program

- ☐
- ☐
- C) Students without programs are excluded
- D) If a program has no students, it won't appear
- E) Can still use WHERE clause to filter results

Answers: A, B, D, E

Question 28

Which are true about PRIMARY KEY constraint? **(Select all that apply)**

- A) Automatically creates an index
- B) Allows NULL values
- C) Ensures uniqueness
- D) A table can have multiple primary keys
- E) Can be composite (multiple columns)

Answers: A, C, E

Question 29

For the EXISTS operator: **(Select all that apply)**

- A) Returns TRUE if subquery returns at least one row
- B) Can be used with correlated subqueries
- C) Is equivalent to COUNT(*) > 0
- D) More efficient than IN for large subqueries
- E) NOT EXISTS checks for absence of rows

Answers: A, B, C, D, E

Question 30

Which require GROUP BY clause? **(Select all that apply)**

- A) Using HAVING clause
- B) Computing aggregate per group
- C) Using aggregate functions

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D) Combining aggregates with non-aggregated columns in SELECT

E) Using ORDER BY

Answers: A, B, D

SECTION C: SCENARIO-BASED QUESTIONS Question 31

Scenario: Database has Student(sid, name, major, gpa) and Enrollment(sid, courseId, grade).

Find students who have enrolled in ALL courses (assuming Course table has 10 courses with courseId 1-10).

Which query is correct?

- A) SELECT sid FROM Enrollment GROUP BY sid HAVING COUNT(*) = 10;
- B) SELECT sid FROM Enrollment GROUP BY sid HAVING COUNT(DISTINCT courseId) = 10;
- C) SELECT sid FROM Student WHERE sid IN (SELECT sid FROM Enrollment);
- D) SELECT DISTINCT sid FROM Enrollment; **Answer: B**

Question 32

Scenario: Employee(empId, name, salary, deptId, managerId) where managerId references empId (self-referencing).

Find employees earning more than their manager.

Which query is correct?

- A) SELECT e.name FROM Employee e WHERE e.salary > (SELECT m.salary FROM Employee m WHERE m.empId = e.managerId);
- B) SELECT e.name FROM Employee e, Employee m WHERE e.managerId = m.empId AND e.salary > m.salary;
- C) Both A and B
- D) Neither works

Answer: C

Question 33

Scenario: Product(pid, name, category, price), Sales(pid, quantity, saleDate).

Find products that have never been sold.

Which query is correct?

- A) SELECT name FROM Product WHERE pid NOT IN (SELECT pid FROM Sales);
- B) SELECT name FROM Product p WHERE NOT EXISTS (SELECT * FROM Sales s WHERE s.pid = p.pid);
- C) SELECT p.name FROM Product p LEFT OUTER JOIN Sales s ON p.pid = s.pid WHERE s.pid IS NULL;
- D) All of the above

Answer: D

Question 34

Scenario: Given: CREATE

TABLE Orders (orderId

INT PRIMARY KEY, custId

INT,

amount DECIMAL(10,2) CHECK (amount > 0),

orderDate DATE DEFAULT CURRENT_DATE,

FOREIGN KEY (custId) REFERENCES Customer(custId) ON DELETE CASCADE

);

What happens when a customer with existing orders is deleted from Customer table?

- A) Deletion fails due to referential integrity
- B) All their orders are automatically deleted
- C) Orders remain with custId set to NULL
- D) Orders remain unchanged

Answer: B

Question 35

Scenario: Student(sid, name, gpa), Enrollment(sid, courseId, semester), Course(courseId, courseName, credits).

Find students who enrolled in more courses in Fall 2024 than Spring 2024.

Which query is correct?

- A)

```
SELECT sid FROM Enrollment
```

```
WHERE semester = 'Fall2024'
```

```
GROUP BY sid
```

```
HAVING COUNT(*) > (SELECT COUNT(*) FROM Enrollment WHERE semester =  
'Spring2024');
```

- B)

```
SELECT e1.sid
```

```
FROM Enrollment e1
```

```
WHERE e1.semester = 'Fall2024'
```

```
GROUP BY e1.sid
```

```
HAVING COUNT(*) > (
```

```
SELECT COUNT(*)  
FROM Enrollment e2  
WHERE e2.sid = e1.sid AND e2.semester = 'Spring2024'  
);
```

- C) Both are correct
- D) Neither is correct

Answer: B

Question 36

Scenario: Employee(empId, name, salary, deptId), Department(deptId, deptName, budget).

Find departments where average employee salary exceeds 25% of department budget.

Which query is correct?

- A)

```
SELECT d.deptName  
FROM Employee e JOIN Department d ON e.deptId = d.deptId  
WHERE AVG(e.salary) > d.budget * 0.25;
```

- B)

```
SELECT d.deptName  
FROM Employee e JOIN Department d ON e.deptId = d.deptId  
GROUP BY d.deptId, d.deptName, d.budget  
HAVING AVG(e.salary) > d.budget * 0.25;
```

- C)

```
SELECT deptName FROM Department  
WHERE budget * 0.25 < (SELECT AVG(salary) FROM Employee WHERE deptId =  
Department.deptId);
```

- D) Both B and C

Answer: D

Question 37

Scenario: Course(courseId, courseName, credits), Prerequisite(courseId, prereqCourseId).

Find courses that have no prerequisites.

Which query is most efficient?

- A) SELECT courseName FROM Course WHERE courseId NOT IN (SELECT courseId FROM Prerequisite);
- B) SELECT courseName FROM Course c WHERE NOT EXISTS (SELECT * FROM Prerequisite p WHERE p.courseId = c.courseId);
- C) SELECT c.courseName FROM Course c LEFT JOIN Prerequisite p ON c.courseId = p.courseId WHERE p.courseId IS NULL;
- D) All have similar efficiency

Answer: B

Question 38

Scenario: Sales(saleId, productId, quantity, price, saleDate).

Find the total revenue per product for sales in 2024, showing only products with revenue > 100000, ordered by revenue descending.

Which query is correct?

- A)

```
SELECT productId, SUM(quantity * price) as revenue
FROM Sales
```

```
WHERE YEAR(saleDate) = 2024 AND SUM(quantity * price) > 100000
GROUP BY productId
ORDER BY revenue DESC;
```

- B)

```
SELECT productId, SUM(quantity * price) as revenue
FROM Sales
WHERE YEAR(saleDate) = 2024
GROUP BY productId
HAVING SUM(quantity * price) > 100000 ORDER
BY revenue DESC;
```

- C) Both work equally
- D) Neither works

Answer: B

Question 39

Scenario: Student(sid, name), Enrollment(sid, courseId, grade), Course(courseId, courseName).

Find students who got 'A' in ALL courses they enrolled in.

Which query is correct?

- A)

```
SELECT s.name
FROM Student s
WHERE NOT EXISTS (
    SELECT * FROM Enrollment e
    WHERE e.sid = s.sid AND e.grade != 'A'
);
```

- B)

```
SELECT s.name
FROM Student s JOIN Enrollment e ON s.sid = e.sid
GROUP BY s.sid, s.name
HAVING COUNT(*) = COUNT(CASE WHEN grade = 'A' THEN 1 END);
```

- C) Both are correct
- D) A is correct only

Answer: C

Question 40

Scenario: Library database: Book(isbn, title, author), Copy(copyId, isbn, status), Loan(loanId, copyId, memberId, loanDate, returnDate).

Find books where ALL copies are currently on loan (returnDate IS NULL).

Which query is correct?

- A)

```
SELECT b.title
```

```
FROM Book b
```

```
WHERE NOT EXISTS (
```

```
    SELECT * FROM Copy c
```

```
    LEFT JOIN Loan l ON c.copyId = l.copyId AND l.returnDate IS NULL
```

```
WHERE c.isbn = b.isbn AND l.loanId IS NULL
```

```
);
```

- B)

```
SELECT b.title
```

```
FROM Book b JOIN Copy c ON b.isbn = c.isbn
```

```
GROUP BY b.isbn, b.title
```

```
HAVING COUNT(c.copyId) = COUNT(CASE WHEN EXISTS (
```

```
    SELECT * FROM Loan l WHERE l.copyId = c.copyId AND l.returnDate IS NULL )
```

```
THEN 1 END);
```

- C) Both work
- D) Neither is fully correct

Answer: A

Question 41

Consider the following tables:

Patient(pid, pname, age)

Admission(pid, admissionDate)

- pid attribute in the Admission table is a foreign key referring to the pid attribute of the Patient table.
- Assume no null values and no foreign keys or integrity constraints.

Given the following queries:

Query1:

```
SELECT pid FROM Patient;
```

Query2:

```
SELECT pid FROM Patient
WHERE pid IN (SELECT pid FROM Admission);
```

Query3:

```
SELECT p.pid
FROM Patient p, Admission a WHERE a.pid = p.pid;
```

Query4:

```
SELECT pid FROM Admission
WHERE EXISTS (
SELECT * FROM Patient
WHERE Patient.pid = Admission.pid
);
```

Question: Which one of the following statements is correct?

- A) Query2 and Query4 return identical row sets, but Query1 and Query2 return different row sets.
- B) All queries return identical row sets.
- C) Query3 returns strictly fewer rows than Query2.
- D) Query4 will encounter an integrity violation at runtime.

Answer: A

Question 42

Consider the following relations:

Product(P_code, Description, Stocking_date, QtyOnHand, MinQty, Price, Discount, V_code)

Vendor(V_code, Name, Address, Phone)

- A vendor can supply more than one product, but a product is supplied by only one vendor.

Question: Which of the following SQL queries will list the Name, Address, and Phone of the vendors who are currently not supplying any product?

- A) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Product);
- B) SELECT Name, Address, Phone FROM Vendor WHERE V_code != Product;
- C) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Vendor);
- D) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Product);
- E) SELECT * FROM Vendor WHERE V_code != P_code;

Answer: A

Question 43

Consider the following table:

Emp (eid, ename, designation, salary, did)

Consider the following relational query on the Emp table:

```
SELECT salary
FROM Emp e1
WHERE 2 = (
SELECT COUNT(DISTINCT(e2.salary))
FROM Emp e2
WHERE e2.salary > e1.salary
);
```

Which one of the following is the correct interpretation of the above query?

- A) Find the 2nd highest salary from the table
- B) Find the 3rd highest salary from the table
- C) Find the 4th highest salary from the table
- D) Find the highest salary from the table

Answer: B

Question 44

Consider the following relation:

Weather (CityID, city, temperature, condition)

Which of the following queries will return the names of those cities with their temperature and condition whose condition is neither 'sunny' nor 'cloudy'?

- A) SELECT city, temperature, condition FROM weather WHERE condition NOT EXISTS ('sunny', 'cloudy');
- B) SELECT city, temperature, condition FROM weather WHERE condition NOT IN ('sunny', 'cloudy');
- C) SELECT city, temperature, condition FROM weather WHERE condition BETWEEN ('sunny', 'cloudy');
- D) SELECT city, temperature, condition FROM weather WHERE condition IN ('sunny', 'cloudy');
- E) SELECT city, temperature, condition FROM weather WHERE condition EXISTS ('sunny', 'cloudy');

Answer: B

Question 45

Consider the following schema:

EMP (eid, ename, age, salary, did)

What will be the output of the following SQL query?

```
SELECT *
FROM emp e
WHERE 2 = (
SELECT COUNT(DISTINCT e1.age)
```

```
FROM emp e1
WHERE e1.age > e.age
);
```

- A) Third highest age
- B) Two distinct ages of employees
- C) Employee with second highest age
- D) Second highest age

Answer: A

Question 46

Which of the following queries will result in an error? (Select one or more)

- A) SELECT dept_id, COUNT(name) FROM employees;
- B) SELECT AVG(salary) FROM employees GROUP BY dept_id;
- C) SELECT eid FROM employees WHERE salary > AVG(salary);
- D) SELECT dept_id, job_id, AVG(salary) FROM employees GROUP BY dept_id, job_id;
- E) SELECT dept_id, AVG(salary) FROM employees GROUP BY dept_id;

Answer: A, C

Question 47

Consider the following tables:

Student (sid, sname, age)

Grades (sid, cid, grade)

- The Student table stores information about all students.
- The Grades table contains the grades that students have obtained for the courses they have completed.

Which of the following SQL queries will produce the names of students who have not completed any courses?

- A) None of the above
- B) SELECT sname FROM Student WHERE sid NOT IN (SELECT sid IN Grades);
- C) SELECT s.sname FROM Student s, Grades g WHERE s.sid = g.sid GROUP BY s.sid HAVING COUNT(*) = 0;
- D) SELECT s.sname FROM Student s WHERE NOT EXISTS (SELECT * FROM Grades g WHERE g.sid = s.sid);
- E) SELECT s.sname FROM Student s LEFT OUTER JOIN Grades g ON s.sid = g.sid WHERE g.sid IS NULL;

Answer: D, E

Question 48

Consider the following relational schema:

Doctor (did, name, specialization)

Patient (pid, pname, docId, diagnosis)

- docId is a foreign key referring to did in the Doctor relation.
- Assume that each doctor diagnoses at least one patient.

Query:

```
SELECT name
FROM Doctor d
WHERE NOT EXISTS (
  SELECT pname
  FROM Patient p
  WHERE p.docId = d.did AND p.diagnosis <> 'Fever'
);
```

- A) Names of all the doctors with all their patients having fever
- B) Names of all the doctors with at least one of their patients having fever
- C) Names of all the doctors with none of their patients having fever
- D) Names of all the doctors with at most one of their patients having fever

Answer: A

Question 49

Consider the following three tables to store student enrollments in different courses:

Student(EnrollNo, Name)

Course(CourseID, Name)

Enrollments(EnrollNo, CourseID)

What does the following query do?

```
SELECT S.Name
FROM Student S, Course C, Enrollments E
WHERE S.EnrollNo = E.EnrollNo
AND C.Name = 'DBMS'
AND E.CourseID = C.CourseID
AND S.EnrollNo IN (
  SELECT S2.EnrollNo
  FROM Student S2, Course C2, Enrollments E2
  WHERE S2.EnrollNo = E2.EnrollNo AND E2.CourseID = C2.CourseID
  AND C2.Name = 'OS'
);
```

- A) Name of all students who are enrolled in "DBMS"
- B) Name of all students who are enrolled in both "DBMS" and "OS"
- C) Name of all students who are either enrolled in "DBMS" or "OS" or both
- D) Name of all students who are either enrolled in "DBMS" or "OS" courses

Answer: B

Question 50

Consider the following table:

Emp(eid, ename, designation, salary, deptName)

Consider the following SQL query:

```
SELECT deptName
```

```
FROM Emp
```

```
WHERE designation = 'Manager'
```

```
GROUP BY deptName
```

```
HAVING AVG(salary) > (SELECT AVG(salary) FROM Emp);
```

Question: This query returns the names of the departments in which:

- A) The average salary of managers is more than the average salary of all male employees in the company
- B) The average salary of managers is more than the average salary in the company
- C) The average salary of managers is more than the average salary of employees in the same department
- D) The average salary is more than the average salary in the company

Answer: B

Question 51

Consider the following relations:

Product(P_code, Description, Stocking_date, QtyOnHand, MinQty, Price, Discount, V_code)

Vendor(V_code, Name, Address, Phone)

- A vendor can supply more than one product, but a product is supplied by only one vendor.

Question: Which of the following SQL queries will list the Name, Address, and Phone of the vendors who are currently not supplying any product?

- A) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Product);
- B) SELECT Name, Address, Phone FROM Vendor WHERE V_code != Product;
- C) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Vendor);
- D) SELECT Name, Address, Phone FROM Vendor WHERE V_code NOT IN (SELECT V_code FROM Product);
- E) SELECT * FROM Vendor WHERE V_code != P_code;

Answer: A

Question 52

Suppose relation R(A, B) currently has tuples {(1,2), (1,3), (3,4)} and relation S(A, C) currently has tuples {(4,6), (7,8)}. Then the number of tuples in the result of the SQL query:

Select * from R left outer join S on (R.B = S.A)

- A) 2
- B) 3
- C) 4
- D) 5

- E) 6

Answer: B

Question 53

Consider the following three tables to store student enrollments in different courses:

Student(EnrollNo, Name)

Course(CourseID, Name) Enrollments(EnrollNo, CourseID) What does the following query do?

```
SELECT S.Name
FROM Student S, Course C, Enrollments E
WHERE S.EnrollNo = E.EnrollNo
AND C.Name = "DBMS"
AND E.CourseID = C.CourseID
AND S.EnrollNo IN (
SELECT S2.EnrollNo
FROM Student S2, Course C2, Enrollments E2
WHERE S2.EnrollNo = E2.EnrollNo
AND E2.CourseID = C2.CourseID
AND C2.Name = "OS"
);
```

- A) Name of all students who are enrolled in "DBMS"
- B) Name of all students who are enrolled in "DBMS" and "OS"
- C) Name of all students who are either enrolled in "DBMS" or "OS" or both
- D) Name of all students who are either enrolled in "DBMS" or "OS" courses

Answer: B

Question 54

Consider the following relational schema: Doctor(did, name, specialization) Patient(pid, pname, docId, diagnosis) docId is a foreign key referring to did of the Doctor relation. Assume that each doctor diagnoses at least one patient.

What does the following query return?

```
SELECT name
FROM Doctor d
WHERE NOT EXISTS (
SELECT pname
FROM Patient p
WHERE p.docId = d.did AND p.diagnosis < 'Fever'
);
```

- A) Names of all the doctors with all their patients having fever
- B) Names of all the doctors with at least one of their patients having fever
- C) Names of all the doctors with none of their patients having fever
- D) Names of all the doctors with at most one of their patient having fever

Answer: A

Question 55

Consider the tables given below:

Student(sid, sname, age)

Grades(sid, cid, grade)

- The Student table stores information of all students.
- The Grades table contains grades the students have obtained for courses he/she had completed.

Which of the following queries would produce the names of the students who had not completed any course? (Select one or more)

- A) None of the above
- B) select sname from Student where sid not IN (select sid in Grades)
- C) select s.sname from Student s, Grades g where s.sid = g.sid group by s.sid having count(*) = 0
- D) select s.sname from Student s where NOT EXISTS (select * from Grades g where g.sid = s.sid)
- E) select s.sname from Student s LEFT OUTER JOIN Grades g where g.sid is NULL

Answer: D, E

Question 56

Suppose the relations are as follows:

R(A, B) with tuples: {(1,2), (1,3), (3,4)}

S(B, C) with tuples: {(2,5), (4,6), (7,8)}

What is the number of tuples in the result of the SQL query: Select * From R Left Outer Join S on (R.B = S.B);

- A) 5
- B) 2
- C) 3
- D) 4
- E) 6

Answer: D

Question 57

Consider a relation Lecturer as follows:

Lecturer(EmpNo CHAR(03), Name VARCHAR(50), Salary REAL, Category VARCHAR(25), DateJoined DATE, DNO)

Consider the following details of a Lecturer:

- EmpNo: 175

- Name: Dulip Silva
- Salary: 12,000
- Category: Instructor
- DateJoined: SYSDATE
- DNO: 05

Which of the following SQL statements will insert the above data into the Lecturer relation? (Select one or more)

- A) INSERT INTO Lecturer (EmpNo, Name, Salary, Category, DateJoined, DNO) VALUES (175, 'Dulip Silva', 12000, 'Instructor', SYSDATE, 05);
- B) INSERT (EmpNo, Name, Salary, Category, DateJoined, DNO) VALUES (175, 'Dulip Silva', 12000, 'Instructor', SYSDATE, 05) INTO Lecturer;
- C) RT INTO Lecturer (EmpNo, Name, Salary, Category, DateJoined, DNO) VALUES (175, 'Dulip Silva', 12000, 'Instructor', SYSDATE, 05);
- D) INSERT INTO Lecturer VALUES (175, 'Dulip Silva', 12000, 'Instructor', SYSDATE, 05);
- E) INSERT INTO Lecturer VALUES (175, 'Dulip Silva', 12000, 'Instructor', SYSDATE, 05);

Answer: A, D, E

Question 58

Which of the following queries will result in an error? (Select one or more)

- A) select avg(salary) from employees group by dept_id
- B) select dept_id, avg(salary) from employees group by dept_id
- C) select eid from employees where salary > avg(salary)
- D) select dept_id, count(name) from employees
- E) select dept_id, job_id, avg(salary) from employees group by dept_id, job_id

Answer: C, D

Question 59

SQL – Avg salary

Emp(eid, ename, designation, salary, deptName)

SELECT deptName

FROM Emp

WHERE designation = 'Manager'

GROUP BY deptName

HAVING AVG(salary) > (SELECT AVG(salary) FROM Emp) The query returns departments where:

- A) Average salary is more than average salary in company
- B) Average salary of managers > average salary in company
- C) Average salary of managers > average salary of employees in same department
- D) Average salary of managers > average salary of all male employees

Answer: B

Question 60

Consider the Appointments table shown below:

Patient	Doctor	AppointmentDate
Lakmal	Dr. Janaka	08-01-2020
Nishani	Dr. Sunila	10-01-2020
Bhagya	Dr. Janaka	07-01-2020

What will be the output of the following SQL query?

```
SELECT COUNT(*)
```

```
FROM
```

```
(
```

```
  (SELECT Patient, Doctor
```

```
    FROM Appointments) AS S
```

```
  INNER JOIN
```

```
    (SELECT Doctor, appointmentDate
```

```
      FROM Appointments) AS T
```

```
);
```

- A) 3

- B) 9

- C) 5

- D) 6

Answer: B

Question 61

Consider the following tables:

Patient(pid, pname, age)

Admission(pid, admissionDate)

- The pid attribute in the Admission table is a foreign key referring to the pid attribute of the Patient table.

- Assume there are no null values or integrity constraint violations.

Given the following four queries:

Query 1: select pid from Patient

Query 2: select pid from Patient where pid in (select pid from Admission)

Query 3: select pid from Admission where pid in (select pid from Patient)

Query 4: select p.pid from Patient p, Admission a where a.pid = p.pid

Query 5: (Alternative representation of Query 4) select pid from Admission where exists (select * from Patient where Patient.pid = Admission.pid)

Which one of the following statements is correct?

- A) Query 2 and Query 4 return identical row sets but Query 1 and Query 2 return different row sets

- B) All queries return identical row sets

- C) Query 3 returns strictly fewer rows than Query 2
- D) Query 4 will encounter an integrity violation at runtime

Answer: A

Question 62

Consider the following schema:

EMP(eid, ename, age, salary, did)

What will be the output of the following SQL query? `select * from emp e where 2 = (select count(distinct e1.age) from emp e1 where e1.age > e.age)`

- A) Third highest age
- B) Two distinct ages of employees
- C) Employee with second highest age
- D) Second highest age

Answer: A

Question 63

Consider the following relation:

Weather(CityID, temperature, city, condition)

Which of the following queries will return the names of cities with temperature and condition whose condition is neither sunny nor cloudy?

- A) `SELECT city, temperature, condition FROM weather WHERE condition NOT EXISTS ('sunny', 'cloudy');`
- B) `SELECT city, temperature, condition FROM weather WHERE condition NOT IN ('sunny', 'cloudy');`
- C) `SELECT city, temperature, condition FROM weather WHERE condition BETWEEN ('sunny', 'cloudy');`
- D) `SELECT city, temperature, condition FROM weather WHERE condition IN ('sunny', 'cloudy');`
- E) `SELECT city, temperature, condition FROM weather WHERE condition EXISTS ('sunny', 'cloudy');`

Answer: B

Question 64

Consider the following table:

Emp(eid, ename, designation, salary, did)

Consider the following relational query on the Emp table:

`SELECT salary`

`FROM Emp e1`

`WHERE 2 = (`

`SELECT COUNT(DISTINCT e2.salary)`

```
FROM Emp e2
WHERE e2.salary > e1.salary
);
```

Which one of the following is the correct interpretation of the above query?

- A) find the 2nd highest salary from the table
- B) find the 3rd highest salary from the table
- C) find the 4th highest salary from the table
- D) find the highest salary from the table

Answer: B

Question 65

SQL query – LEFT OUTER JOIN

Relations $R(A, B) = \{(1,2), (3,1), (3,4)\}$

$S(A, B) = \{(2,3), (1,4), (7,8)\}$

SQL: SELECT * FROM R LEFT OUTER JOIN S ON R.B = S.A

Number of tuples in result:

- A) 4
- B) 5
- C) 2
- D) 6
- E) 3

Answer: E

Question 66

SQL – Employee hierarchy

Emps(empID, ssNo, name, mgrID)

Find employees managed by people managed by employee ID 123:

SELECT ee.empID

FROM Emps ee, Emps ff

WHERE ee.mgrID = ff.empID AND ff.mgrID = 123;2.

SELECT empID

FROM Emps

WHERE mgrID IN (SELECT empID FROM Emps WHERE mgrID = 123);

Which query is correct?

- A) Query can't represent in SQL
- B) Both I and II
- C) I only
- D) Neither I nor II
- E) II only

Answer: B

Question 67

Consider the following table:

Emp(eid, ename, designation, salary, did) SQL query:

```
SELECT salary
FROM Emp e1
WHERE 2 = (
SELECT COUNT(DISTINCT(e2.salary))
FROM Emp e2 WHERE e2.salary > e1.salary
)
```

Which one of the following is the correct interpretation of the above query?

- A) Find the 2nd highest salary from table
- B) Find the 4th highest salary from table
- C) Find the highest salary from table
- D) Find the 3rd highest salary from table

Answer: D

Question 68

SQL – 2nd highest salary

Emp(eid, ename, designation, salary, did)

```
SELECT salary
FROM Emp e1
WHERE 2 = (
SELECT COUNT(DISTINCT(e2.salary))
FROM Emp e2
WHERE e2.salary > e1.salary
)
```

Correct interpretation:

- A) Find the 2nd highest salary
- B) Find the 4th highest salary
- C) Find the highest salary
- D) Find the 3rd highest salary

Answer: D

Question 69

Relation Emps(empID, ssNo, name, mgrID)

Suppose we wish to find the IDs of the employees that are managed by people who are managed by the employee with ID 123. Queries:

1.

```
SELECT ee.empID
```

FROM Emps ee, Emps ff
WHERE ee.mgrID = ff.empID AND ff.mgrID = 123;

2. SELECT empID
FROM Emps
WHERE mgrID IN
(SELECT empID FROM Emps WHERE mgrID = 123);

Which, if any, of the two queries above will correctly get the desired set of employee IDs?

- A) Query can't represent in SQL
- B) Both I and II
- C) I only
- D) Neither I nor II
- E) II only

Answer: B

Question 70

Weather(CityID, Temperature, City, Condition)

Which query will return the names of cities with temperature and condition where condition is neither 'sunny' nor 'cloudy'?

- A) SELECT city, temperature, condition FROM weather WHERE condition EXISTS ('sunny', 'cloudy');
- B) SELECT city, temperature, condition FROM weather WHERE condition NOT IN ('sunny', 'cloudy');
- C) SELECT city, temperature, condition FROM weather WHERE condition IN ('sunny', 'cloudy');
- D) SELECT city, temperature, condition FROM weather WHERE condition BETWEEN ('sunny', 'cloudy');
- E) SELECT city, temperature, condition FROM weather WHERE condition NOT EXISTS ('sunny', 'cloudy');

Answer: B

Question 71

The DROP TABLE statement: (Select one or more)

- A) Deletes the table structure only.
- B) Works whether or not referential integrity constraints are violated.
- C) Deletes the table structure along with the table data.
- D) Works only if referential integrity constraints are not violated.
- E) Deletes the table data only.

Answer: C, D

Question 72

Consider tables:

- Patient(pid, pname)
- Admission(pid, admissionDate) pid in Admission is a foreign key referencing pid in Patient. No nulls exist.

Given queries:

1. SELECT pid FROM Patient WHERE pid IN (SELECT pid FROM Admission);
2. SELECT pid FROM Admission WHERE pid IN (SELECT pid FROM Patient);
3. SELECT p.pid FROM Patient p, Admission a WHERE a.pid = p.pid;
4. SELECT pid FROM Admission WHERE EXISTS (SELECT * FROM Patient WHERE Patient.pid = Admission.pid);

Which statement is correct?

- A) All queries return identical row sets.
- B) Query 4 will encounter an integrity violation at runtime.
- C) Query 3 returns strictly fewer rows than Query 2.
- D) Query 2 and Query 4 return identical row sets but Query 1 and Query 2 return different row sets.

Answer: D

Question 73

Which of the following statements are not true about ALTER and UPDATE statements? (Select one or more)

- A) ALTER TABLE command could be used to modify the structure of the table.
- B) UPDATE command could be used to modify the structure of the table.
- C) UPDATE command could be used to edit values of rows in a table.
- D) ALTER TABLE command could be used to edit values of rows in a table.
- E) UPDATE command could be used to add constraints after table creation.

Answer: B, D, E

Question 74

Which of the following queries will result in an error? (Select one or more)

- A) SELECT dept_id, AVG(salary) FROM employees GROUP BY dept_id;
- B) SELECT dept_id, COUNT(name) FROM employees;
- C) SELECT AVG(salary) FROM employees GROUP BY dept_id;
- D) SELECT dept_id, job_id, AVG(salary) FROM employees GROUP BY dept_id, job_id;
- E) SELECT eid FROM employees WHERE salary > AVG(salary);

Answer: B, E

Question 75

Which of the following statements are not true about ALTER and UPDATE statements? (Select one or more)

- A) UPDATE command could be used to modify the structure of the table
- B) UPDATE command could be used to add constraints after table creation
- C) ALTER TABLE command could be used to edit values of rows in a table
- D) ALTER TABLE command could be used to modify the structure of the table

Answer: A, B, C

Question 76

ALTER and UPDATE statements Which statements are not true?

- A) ALTER TABLE command could be used to edit values of the rows in a table
- B) ALTER TABLE command could be used to modify the structure of the table
- C) UPDATE command could be used to edit values of the rows in a table
- D) UPDATE command could be used to add constraints to a table after it is created
- E) UPDATE command could be used to modify the structure of the table

Answer: A, D, E

Question 77

Suppose relations:

- $R(A, B) = \{(1,2), (1,3), (3,4)\}$
- $S(B, C) = \{(2,5), (4,6), (7,8)\}$

What is the number of tuples in the result of:

SELECT *

FROM R LEFT OUTER JOIN S ON (R.B = S.B);

- A) 2
- B) 3
- C) 4
- D) 5
- E) 6

Answer: B

Question 78

Which of the following statements are not true about ALTER and UPDATE statements?

- A) ALTER TABLE command could be used to edit values of the rows in a table
- B) ALTER TABLE command could be used to modify the structure of the table
- C) UPDATE command could be used to edit values of the rows in a table
- D) UPDATE command could be used to add constraints to a table after it is created
- E) UPDATE command could be used to modify the structure of the table

Answer: A, D, E

Question 79

Student (sid, sname, age), Grades(sid, cid, grade) — Grades table stores grades obtained.

Which queries would produce the names of students who have not completed any course? (Select one or more)

- A) SELECT s.sname FROM Student s LEFT OUTER JOIN Grades g ON s.sid = g.sid WHERE g.sid IS NULL;
- B) SELECT sname FROM Student WHERE sid NOT IN (SELECT sid FROM Grades);
- C) SELECT s.sname FROM Student s, Grades g WHERE s.sid = g.sid GROUP BY s.sid HAVING COUNT(*) = 0;
- D) SELECT s.sname FROM Student s WHERE NOT EXISTS (SELECT * FROM Grades g WHERE g.sid = s.sid);
- E) None of the above

Answer: A, B, D

Question 80

Given tables:

Member(memid, name, address, phone)

Member_Hobbies(memid, hobby)

Which of the following queries will return the names of members who have both 'Music' and 'Photography' as hobbies? (Select one or more)

- A) SELECT m.name FROM Member m, Member_Hobbies h WHERE m.memid = h.memid AND h.hobby IN ('Music', 'Photography')
- B) SELECT m.name FROM Member m WHERE m.memid IN (SELECT memid FROM Member_Hobbies WHERE hobby = 'Music') AND m.memid IN (SELECT memid FROM Member_Hobbies WHERE hobby = 'Photography')
- C) SELECT m.name FROM Member m, Member_Hobbies h WHERE m.memid = h.memid AND h.hobby = ANY ('Music', 'Photography')
- D) SELECT m.name FROM Member m, Member_Hobbies h WHERE m.memid = h.memid AND (h.hobby = 'Music' AND h.hobby = 'Photography')
- E) SELECT m.name FROM Member m, Member_Hobbies h WHERE m.memid = h.memid AND m.memid IN (SELECT memid FROM Member_Hobbies WHERE hobby = 'Photography') AND m.memid IN (SELECT memid FROM Member_Hobbies WHERE hobby = 'Music');

Answer: B, E

Question 81

Consider Emp(eid, ename, designation, salary, deptName). What does this SQL return?
SELECT deptName FROM Emp WHERE designation = 'Manager' GROUP BY deptName HAVING
AVG(salary) > (SELECT AVG(salary) FROM Emp); It returns the names of the departments in
which:

- A) the average salary of managers is more than the average salary of all male employees in the company.
- B) the average salary of managers is more than the average salary in the company.
- C) the average salary is more than the average salary in the company.
- D) the average salary of managers is more than the average salary of employees in the same department.

Answer: B

Question 82

Three tables to store student enrollments in courses:

- Student(EnrollNo, Name)
- Course(CourseID, Name)
- Enrollments(EnrollNo, CourseID) What does the following query do?
SELECT S.Name FROM Student S, Course C, Enrollments E WHERE S.EnrollNo = E.EnrollNo
AND C.Name = 'DBMS' AND E.CourseID = C.CourseID AND S.EnrollNo IN (SELECT S2.EnrollNo
FROM Student S2, Course C2, Enrollments E2 WHERE S2.EnrollNo = E2.EnrollNo AND C2.Name
= 'OS' AND E2.CourseID = C2.CourseID);
- A) Names of all students enrolled in either "DBMS" or "OS".
- B) Names of all students enrolled in "DBMS" and "OS".
- C) Names of all students enrolled in "DBMS" only.
- D) Names of all students enrolled in "OS" only.

Answer: B

Question 83

Consider Emp(eid, ename, designation, salary, did). What does this query return?
SELECT salary FROM Emp e1 WHERE 2 = (SELECT COUNT(DISTINCT(e2.salary)) FROM Emp
e2 WHERE e2.salary > e1.salary);

- A) 2nd highest salary from table
- B) 3rd highest salary from table
- C) Highest salary from table
- D) 4th highest salary from table

Answer: B

Question 84

SQL – Cities with higher temperature than Paris Weather(CityID, Temperature, City, Condition)

Correct query:

- A) SELECT temperature > MIN(temperature) FROM weather WHERE city='Paris' GROUP BY city
- B) SELECT city FROM weather WHERE city='Paris' AND temperature > ALL (temperature)
- C) SELECT city FROM weather WHERE temperature > (SELECT temperature FROM weather WHERE city='Paris')
- D) SELECT city FROM weather WHERE temperature EXISTS (SELECT FROM weather WHERE city='Paris')
- E) SELECT city FROM weather GROUP BY Paris

Answer: C

Question 85

Student enrollments query

Tables: Student(EnrollNo, Name), Course(CourseID, Name), Enrollments(EnrollNo, CourseID)

Query:

SELECT S.Name

FROM Student S, Course C, Enrollments E

WHERE S.EnrollNo = E.EnrollNo AND C.Name = "DBMS" AND E.CourseID = C.CourseID

AND S.EnrollNo IN (SELECT S2.EnrollNo

FROM Student S2, Course C2, Enrollments E2

WHERE S2.EnrollNo = E2.EnrollNo AND C2.Name = "OS" AND E2.CourseID = C2.CourseID

)

Result:

- A) Name of all students
- B) Name of all students who are either enrolled in DBMS or OS or both
- C) Name of all students enrolled in DBMS
- D) Name of all students enrolled in DBMS and OS

Answer: D

Question 86

Consider the following relational scheme with all atomic values:

Academic_Staff(SID, FacultyID, FacultyLocation, FacultyPhone, StaffName, StaffPosition, HoursPerWeek)

With the following functional dependencies:

1. SID → StaffName, StaffPosition, FacultyID, FacultyLocation, FacultyPhone

2. FacultyID $\rightarrow$ FacultyLocation, FacultyPhone
3. FacultyLocation $\rightarrow\rightarrow$ FacultyID, FacultyPhone
4. FacultyPhone $\rightarrow$ FacultyID, FacultyLocation Question: What is the primary key for the relation?
- A) FacultyLocation
 - B) SID
 - C) FacultyPhone
 - D) StaffName
 - E) SID, FacultyID

Answer: B

Question 87

Suppose relation R(A, B) currently has tuples: { (1,2), (1,3), (3,4) } and relation S(B, C) currently has tuples: { (2,5), (4,6), (7,8) }

What is the number of tuples in the result of the following SQL query?

```
SELECT *  
FROM R  
LEFT OUTER JOIN S ON R.B = S.B;
```

- A) 5
- B) 2
- C) 3
- D) 4
- E) 6

Answer: D

Question 88

Which of the following queries will result in an error? (Select one or more)

- A) SELECT dept_id, COUNT(name) FROM employees;
- B) SELECT eid FROM employees WHERE salary > AVG(salary);
- C) SELECT dept_id, AVG(salary) FROM employees GROUP BY dept_id;
- D) SELECT dept_id.job_id, AVG(salary) FROM employees GROUP BY dept_id.job_id;
- E) SELECT AVG(salary) FROM employee GROUP BY dept_id;

Answer: A, B, D

Question 89

In the context of SQL as a comprehensive language, which components beyond DDL and DML are highlighted in the review? (Select one or more)

- A) Facilities for security and authorization
- B) Facilities for transaction processing

- C) Facilities for embedding SQL in general-purpose languages
- D) Facilities for physical storage optimization

Answer: A, B, C

Question 90

What is the effect of specifying ON DELETE SET DEFAULT in a FOREIGN KEY constraint, as in the STUDENT table example?

- A) It cascades deletions to referencing rows
- B) It sets the foreign key to NULL on deletion
- C) It sets the foreign key to its default value on deletion
- D) It prevents deletion if references exist

Answer: C

Question 91

In the CREATE TABLE STUDENT example, why might an INSERT fail for a row with progId not in PROGRAMS? (Select one or more)

- A) Due to the FOREIGN KEY constraint
- B) Because progId has DEFAULT 'IT'
- C) If ON UPDATE CASCADE is violated
- D) The UNIQUE on nic prevents it

Answer: A

Question 92

Which ALTER TABLE operations are demonstrated in the lecture? (Select one or more)

- A) ADD COLUMN
- B) ADD CONSTRAINT
- C) DROP COLUMN
- D) DROP TABLE

Answer: A, B, C

Question 93

How does DROP TABLE differ from ALTER TABLE DROP COLUMN in impact?

- A) DROP TABLE removes the entire table; ALTER DROP COLUMN removes one column
- B) ALTER DROP COLUMN cascades to foreign keys; DROP TABLE does not
- C) DROP TABLE preserves data; ALTER DROP COLUMN does not
- D) Both require no references to succeed

Answer: A

Question 94

In INSERT statements, what happens if columns are omitted, as in the second STUDENT insert example?

- A) Omitted columns get NULL if allowed
- B) Omitted columns get DEFAULT if specified
- C) The insert fails if PRIMARY KEY is omitted
- D) All columns must be specified explicitly

Answer: A, B, C

Question 95

For the DELETE statement example, what would happen if WHERE is omitted?

- A) It deletes all rows in the table
- B) It fails due to syntax error
- C) It deletes only NULL rows
- D) It prompts for confirmation

Answer: A

Question 96

In UPDATE without WHERE, what is the risk? (Select one or more)

- A) Updates all rows in the table
- B) May violate CHECK constraints
- C) Could cascade via FOREIGN KEY
- D) Fails if no SET clause

Answer: A, B, C

Question 97

Which is the correct basic form of SELECT as per the lecture?

- A) SELECT <attributes> FROM <relations> WHERE <conditions>
- B) SELECT * WHERE <conditions> FROM <relations>
- C) FROM <relations> SELECT <attributes> WHERE <conditions>
- D) WHERE <conditions> SELECT <attributes> FROM <relations>

Answer: A

Question 98

In the LIKE example for names starting with 'A', what would LIKE 'A_' match?

- A) Names with exactly two characters starting with A
- B) Any name starting with A
- C) Names ending with A
- D) Names containing A anywhere

Answer: A

Question 99

Why is IS NULL used instead of = NULL in WHERE clauses?

- A) = NULL always returns false; IS NULL checks properly
- B) IS NULL is for strings only
- C) = NULL works for numbers
- D) IS NULL is deprecated

Answer: A

Question 100

What distinguishes INNER JOIN from cartesian product in the example? (Select one or more)

- A) INNER JOIN uses ON or WHERE for condition
- B) Cartesian has no condition, all combinations
- C) INNER JOIN eliminates unmatched rows
- D) Both return same if no condition

Answer: A, B, C

Question 101

In LEFT OUTER JOIN example, why are NULLs in offerBy for some students?

- A) No matching program for that pid
- B) Programs without students are excluded
- C) It's the right table's unmatched
- D) LEFT includes all left rows

Answer: A, C, D

Question 102

In the IN nested query example for 'A' grades, what does the subquery return? (Select one or more)

- A) List of SID with Grade='A'
- B) Matches outer SID to subquery list
- C) Could use EXISTS alternatively
- D) Requires correlated subquery

Answer: A, B, C
